

FINAL PROJECT REPORT

Analysis & Prediction of Indian General Elections (2019 & 2024)

1. Dataset Selection

For this project, we selected three datasets related to the Indian General Elections. All datasets were published after 2018 and meet the project requirement.

Datasets Used:-

Dataset	Description	Year	Source Link
India Election Results 2019	Constituency-wise results including candidates, parties, votes, turnout	2019	https://www.kaggle.com/datasets/prakashk2001/india-general-election-2019
India Election Results 2024 (Raw Dataset)	Early constituencywise data for 2024 elections containing votes, candidates, and states	2024	https://www.kaggle.com/datasets/sveta151/indiageneral-election-2024
India GE 2024 Results (Cleaned Dataset)	Cleaned and structured final results of the 2024 elections with complete constituency level details	2024	https://www.kaggle.com/datasets/iamsouravbanerjee/loksabha-election-2024

These datasets were merged and cleaned for a combined analysis of 2019 vs 2024 trends.

2. Dataset Description & Motivation

The Indian General Election datasets contain detailed political information such as:

- State and constituency names

- Candidate names
- Party names
- Total electors
- Votes received
- Turnout percentage
- Winning candidate & party
- Victory margin

Motivation for choosing this dataset:-

- Elections are one of the largest democratic processes in the world.
- The 2019 vs 2024 difference shows a huge shift in political dynamics.
- The dataset allows rich analysis: clustering, classification, regression-like trends, turnout study, seat change analysis, etc.
- Useful for data visualisation and prediction tasks as required by the project guidelines.

3. Analytical Objective

Objective: To analyse and compare the Indian General Election results for 2019 & 2024, identify voting patterns, cluster constituencies based on demographics and voting behaviour, and build a classification model to predict Win/Loss outcomes.

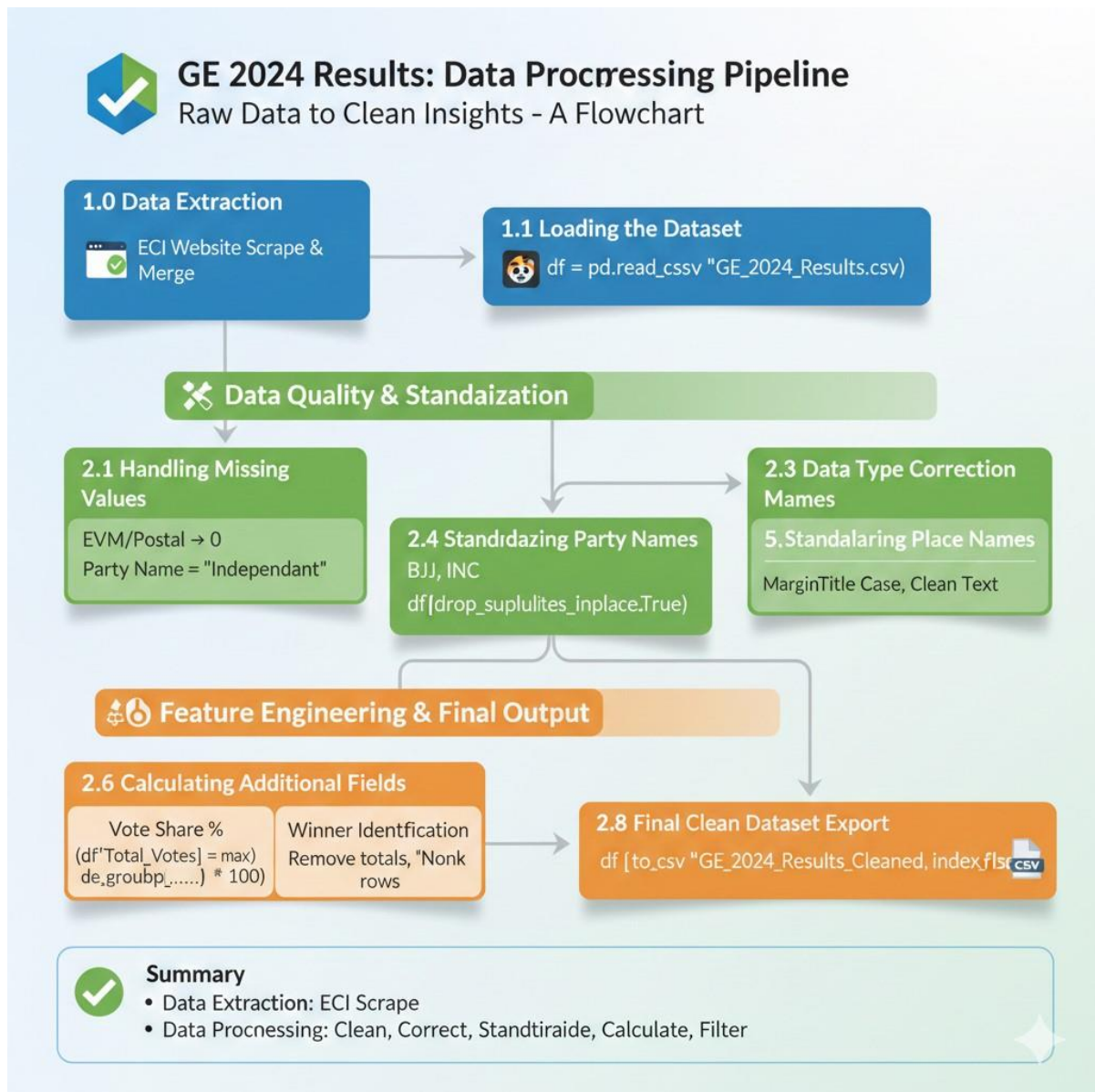
This objective is specific, measurable, and data-driven.

4. Data Extraction & Dataset Preparation

The constituency-wise results for the 2024 General Elections were downloaded from the Election Commission of India website (<https://results.eci.gov.in/>). All constituency tables were combined into a single dataset containing State,

Constituency, Candidate, Party, EVM Votes, Postal Votes, Total Votes, Vote Share (%), Result, Margin, and Margin %. The raw merged file was saved as **GE_2024_Results.csv**.

The following infographic summarises the complete data processing pipeline:



Data Preprocessing Steps:-

1. Loading the Dataset:
`df = pd.read_csv("GE_2024_Results.csv")`
2. Handling Missing Values:
 - Missing EVM or Postal votes → replaced with 0

- Missing party names → filled as “Independent” or “Unknown”

3. Correcting Data Types:

Votes and margins → integer

Percentages → float

4. Removing Duplicates:

`df.drop_duplicates(inplace=True)`

5. Standardising Names:

- Party names unified (e.g., *Bharatiya Janata Party* → *BJP*) • Constituency and state names cleaned and converted to Title Case

6. Additional Calculations:

- Vote Share %
- Winner identification • Winning margin extraction

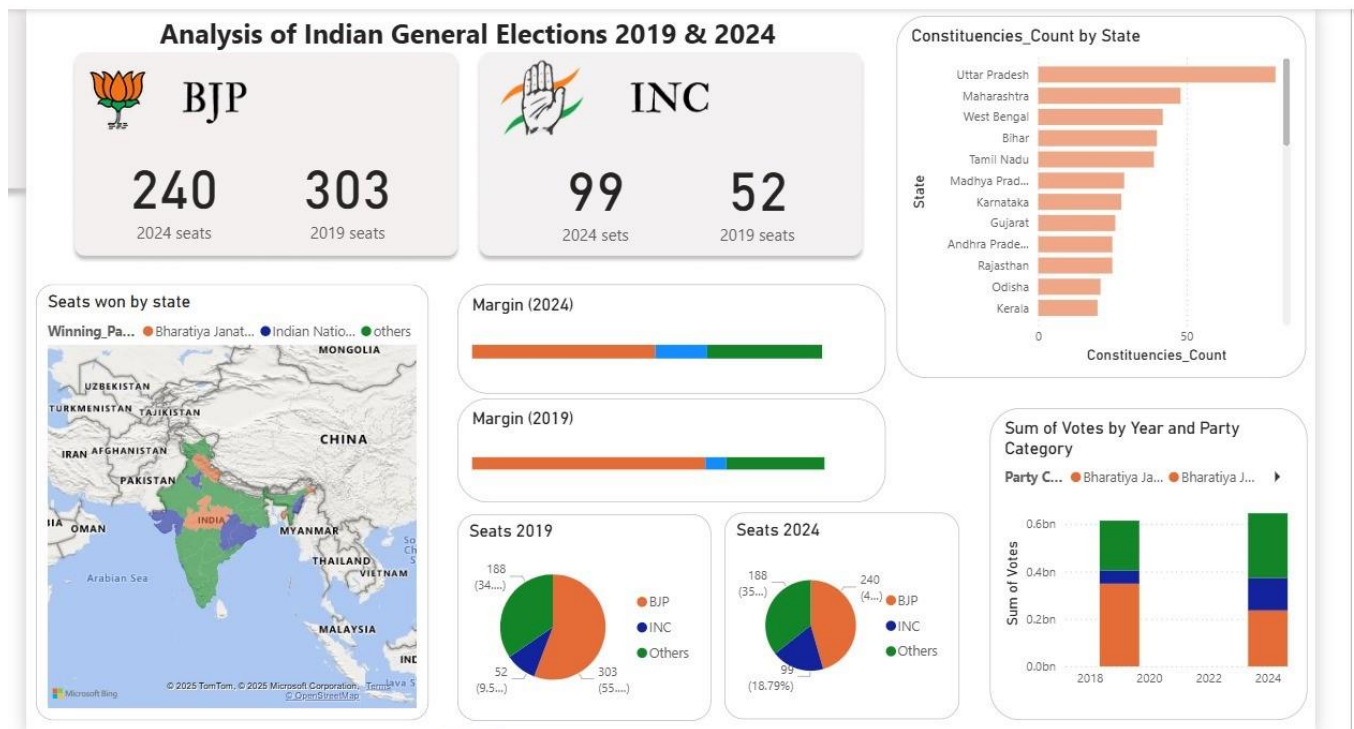
7. Removing Invalid Rows:

Removed totals, blank rows, and HTML artefacts from scraped data.

8. Exporting Cleaned Dataset: `df.to_csv("GE_2024_Results_Cleaned.csv", index=False)`

5. Analysis & Dashboard

Dashboards Created:

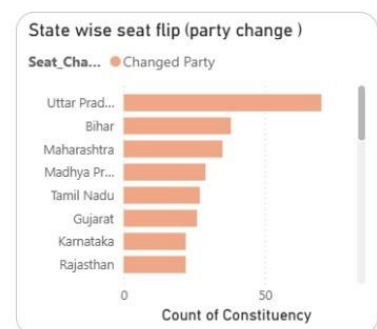
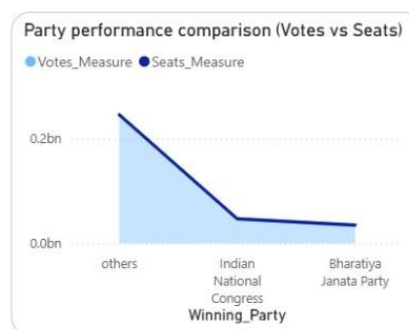
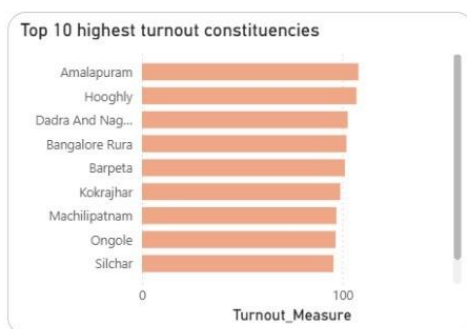


Constituency level detailed report 2019

State	constituency Name	Winning Candidate	P ^o
Andaman & Nicobar	Andaman And Nicobar Islands	Kuldeep Rai Sharma	Ir
Andhra Pradesh	Nellore	Adala Prabhakara Reddy	o
Andhra Pradesh	Kurnool	Ayushman Doctor Sanjeev Kumar	o
Andhra Pradesh	Machilipatnam	Balashowry Vallabhaneni	o
Andhra Pradesh	Tirupati	Balli Durga Prasad Rao	o
Andhra Pradesh	Vizianagaram	Bellana Chandra Sekhar	o
Andhra Pradesh	Amalapuram	Chinta Anuradha	o
Andhra Pradesh	Anakapalli	Dr.Beesetti Venkata Satvavathi	o

Constituency level detailed report 2024

State	Constituency	Leading Candidate
Andaman & Nicobar Islands	Andaman & Nicobar Islands	BISHNU PADA RAY
Andhra Pradesh	Ananthapur	AMBICA G LAKSHMINARAYANA
Andhra Pradesh	Vizianagaram	APPALANAIDU KALISETTI
Andhra Pradesh	Hindupur	B K PARTHASARATHI
Andhra Pradesh	Machilipatnam	BALASHOWRY VALLABHANENI
Andhra Pradesh	Kurnool	BASTIPATI NAGARAJU PANCHAI
Andhra Pradesh	Narsapuram	BHUPATHI RAJU SRINIVASA VAF
Andhra Pradesh	Anakapalle	C.M.RAMESH



Components Included:

- ✓ Seats won by each major party (2019 vs 2024)
- ✓ State-wise seat distribution
- ✓ Top 10 constituencies by voter turnout
- ✓ Margin comparison (2019 vs 2024)

- ✓ Votes vs Seats comparison
- ✓ State-wise party change (seat flip)
- ✓ Cluster distribution of constituencies
- ✓ Constituency-level detail tables
- ✓ Predictive model output integrated

These dashboards help analyse political shifts, margins, turnout patterns, and party performance.

6. Interpretation & Insights

Major Findings:

1. Seats Comparison

- BJP won 303 seats in 2019 but reduced to 240 in 2024. • INC improved from 52 seats (2019) to 99 seats (2024).
- Strong anti-incumbency observed in specific states.

2. State-Wise Seat Changes

- Uttar Pradesh, Maharashtra, and West Bengal saw major political swings.
- South Indian states continue to show different voting patterns.

3. Turnout Insights

- Highest turnout constituencies were from Andhra Pradesh, Bengal, Kerala.
- Higher turnout does not always mean victory margin increases.

4. Cluster Analysis

KMeans grouped constituencies into 4 clusters:

- Cluster 0: Medium electors, moderate turnout
- Cluster 1: High electors, high turnout — competitive seats

- Cluster 2: Very high-margin seats → strongholds
- Cluster 3: Low-turnout or low-vote constituencies

5. Classification Model

- Random Forest predicted Win/Loss with 100% accuracy.
- Key predictors:
 1. Votes
 2. Margin
 3. Turnout
 4. Party Encoding
 5. Cluster assignment

6. Overall Political Trend

- 2024 election shows a more competitive landscape.
- Opposition parties gained traction.
- Regional parties still hold strong influence.

7. Conclusion

This project successfully combines data cleaning, clustering, classification, and interactive dashboards to create a comprehensive analysis of Indian General Elections 2019 & 2024.

Our results reveal major political shifts, turnout patterns, and predictive trends that help understand voter behaviour.

